

1BM24CS299
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5K Batch 2

AI LAB 1

25/9/26

1. Tic Tac Toe

Code:

```
board = [" " for i in range(9)]

while True:

    print(board[0], "|", board[1], "|", board[2])
    print("---+---+---")
    print(board[3], "|", board[4], "|", board[5])
    print("---+---+---")
    print(board[6], "|", board[7], "|", board[8])

    #HUMAN INPUT
    pos = int(input("Enter position (1-9): ")) - 1

    if board[pos] != " ":
        print("Position already occupied!")
        continue

    board[pos] = "X"

    #HUMAN WIN CHECK
    if ((board[0] == board[1] == board[2] == "X") or
        (board[3] == board[4] == board[5] == "X") or
        (board[6] == board[7] == board[8] == "X") or
        (board[0] == board[3] == board[6] == "X") or
        (board[1] == board[4] == board[7] == "X") or
        (board[2] == board[5] == board[8] == "X") or
        (board[0] == board[4] == board[8] == "X") or
        (board[2] == board[4] == board[6] == "X")):

        print("You Win!")
        break

    #DRAW CHECK
    if " " not in board:
        print("Draw!")
        break

    #COMPUTER INPUT
    for i in range(9):
        if board[i] == " ":
            board[i] = "O"
            break

    #COMPUTER WIN CHECK
```

```

if (board[0] == board[1] == board[2] == "O") or
    (board[3] == board[4] == board[5] == "O") or
    (board[6] == board[7] == board[8] == "O") or
    (board[0] == board[3] == board[6] == "O") or
    (board[1] == board[4] == board[7] == "O") or
    (board[2] == board[5] == board[8] == "O") or
    (board[0] == board[4] == board[8] == "O") or
    (board[2] == board[4] == board[6] == "O")):

    print("Computer Wins!")
    break

#DRAW CHECK
if " " not in board:
    print("Draw!")
    break

```

Output:

```

...  |  |
--+--+
    |  |
--+--+
    |  |
Enter position (1-9): 1
X | O |
--+--+
    |  |
--+--+
    |  |
Enter position (1-9): 3
X | O | X
--+--+
O |  |
--+--+
    |  |
Enter position (1-9): 5
X | O | X
--+--+
O | X | O
--+--+
    |  |
Enter position (1-9): 8
X | O | X
--+--+
O | X | O
--+--+
O | X |
Enter position (1-9): 9
You Win!

```

2. Vacuum Cleaner

Code:

```
room_a = input("Enter status of Room A (dirty/clean): ")
room_b = input("Enter status of Room B (dirty/clean): ")

location = input("Enter vacuum location (A/B): ")

print("\nInitial State:")
print("Room A:", room_a)
print("Room B:", room_b)
print("Vacuum Location:", location)

while room_a == "dirty" or room_b == "dirty":

    if location == "A":

        if room_a == "dirty":
            print("Room A is dirty -> Cleaning Room A")
            room_a = "clean"

        else:
            print("Room A is clean -> Moving to Room B")
            location = "B"

    else:

        if room_b == "dirty":
            print("Room B is dirty -> Cleaning Room B")
            room_b = "clean"

        else:
            print("Room B is clean -> Moving to Room A")
            location = "A"

if room_a == "clean" and room_b == "clean":
    print("\nGoal achieved!")
    print("Both rooms are clean.")
```

Output:

```
*** Enter status of Room A (dirty/clean): dirty
Enter status of Room B (dirty/clean): dirty
Enter vacuum location (A/B): A

Initial State:
Room A: dirty
Room B: dirty
Vacuum Location: A
Room A is dirty -> Cleaning Room A
Room A is clean -> Moving to Room B
Room B is dirty -> Cleaning Room B

Goal achieved!
Both rooms are clean.
```